

Lecture 7

Inner Product Spaces

The inner product is taken in the physics convention: conjugate-linear in the first slot, linear in the second, so $\langle \phi | \psi \rangle = \langle \phi, \psi \rangle$ is linear in the ket. Unless stated otherwise, the inner product is the standard one. Tags mark the flavour; a ★ marks a harder one.

Core tutorial route

Work **7.2, 7.1(i), 7.6, 7.8(i), 7.10, and 7.13(iii)** in that order. This route covers the inner-product axioms, computing in $\mathbb{C}^n$, the orthogonal decomposition, a proof of the Cauchy–Schwarz inequality in $\mathbb{C}^2$, the polarization identity, and a quantum amplitude.

Additional practice: 7.1(ii), 7.3–7.5, 7.7(i), 7.8(ii), 7.9, 7.11, 7.13(i)–(ii), 7.15(i). **Bridge:** 7.1(iii), 7.7(ii), 7.14, 7.15(ii) (function spaces, a rank-one projector, and the expansion in an orthonormal basis). **Extension:** 7.12 (the equality case of the triangle inequality).

Inner products, norms, and orthogonality

Exercise 7.1 (computational)

- (i) In $\mathbb{C}^3$ take $u = (1, -i, 2)$ and $v = (i, 1, 1 - i)$. Compute $\langle u, v \rangle$, $\langle v, u \rangle$, $\|u\|$, and $\|v\|$.
- (ii) On $M_{2 \times 2}(\mathbb{R})$ with the Frobenius inner product $\langle A, B \rangle = \text{tr}(A^\dagger B)$, take $A = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$. Compute $\langle A, B \rangle$, $\|A\|$, and $\|B\|$.
- (iii) On $\mathcal{C}[0, 1]$ with $\langle f, g \rangle = \int_0^1 \overline{f} g \, dt$, take $f(t) = 1 + t$ and $g(t) = 1 - t$. Compute $\langle f, g \rangle$, $\|f\|$, and $\|g\|$.

Answer. (i) $\langle u, v \rangle = 2$, $\langle v, u \rangle = 2$, $\|u\| = \sqrt{6}$, $\|v\| = 2$. (ii) $\langle A, B \rangle = 4$, $\|A\| = 3$, $\|B\| = \sqrt{6}$. (iii) $\langle f, g \rangle = \frac{2}{3}$, $\|f\| = \sqrt{7/3}$, $\|g\| = \frac{1}{\sqrt{3}}$.

Exercise 7.2 (proof) On $\mathbb{C}^2$ consider the form $B(u, v) = \overline{u_1}v_1 - \overline{u_2}v_2$. Decide whether B is an inner product in the physics convention. Check conjugate symmetry, linearity in the second slot, and positive definiteness separately.

Answer. B is conjugate-symmetric and linear in the second slot, but is not positive definite: $B((0, 1), (0, 1)) = -1$.

Hint. Two of the axioms can be checked directly. For the third, look for a nonzero vector on which B takes a negative value; try e_2 .

Exercise 7.3 (computational) Find all a such that the two vectors are orthogonal: (i) $(1, a, 2)$ and $(3, -1, 1)$ in $\mathbb{R}^3$, $a \in \mathbb{R}$; (ii) $(1, a)$ and $(1, i)$ in $\mathbb{C}^2$, $a \in \mathbb{C}$.

Answer. (i) $a = 5$. (ii) $a = -i$.

Exercise 7.4 (computational) In $\mathbb{R}^3$ take $v_1 = (1, 1, 0)$ and $v_2 = (0, 1, 1)$. Compute the Gram matrix $G_{jk} = \langle v_j, v_k \rangle$ of these vectors, and use it to decide whether the vectors are linearly independent.

Answer. $G = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\det G = 3 \neq 0$, so the vectors are linearly independent.

Exercise 7.5 (proof)

- (i) Let U be a subspace of an inner product space V . Prove that $U \cap U^\perp = \{0\}$.
- (ii) Let $v_1, \dots, v_m$ be nonzero, pairwise orthogonal vectors. Prove that they are linearly independent.

Hint. (i) A vector lying in both sets is orthogonal to every vector of U , itself included. (ii) Pair the equation $\sum_k c_k v_k = 0$ with each v_j ; by orthogonality only one term survives.

The Cauchy–Schwarz inequality

Exercise 7.6 (computational) In $\mathbb{R}^2$ take $u = (4, 2)$ and $v = (1, 1)$. Split the vector u into a part parallel to v and a part orthogonal to v . Find the parallel part by the formula $\frac{\langle v, u \rangle}{\|v\|^2} v$.

Answer. $u = (3, 3) + (1, -1)$, where $(3, 3) = 3v$ is parallel to v and $(1, -1) \perp v$.

Exercise 7.7 (computational)

- (i) In $\mathbb{C}^2$ take $u = (1, 1)$ and $v = (1, i)$. Compute $|\langle u, v \rangle|$, $\|u\|$, and $\|v\|$, and check the Cauchy–Schwarz inequality.
- (ii) On $\mathcal{C}[0, 1]$ with $\langle f, g \rangle = \int_0^1 \bar{f} g \, dt$, compute $\langle 1, t^2 \rangle$, $\|1\|$, and $\|t^2\|$, and check the Cauchy–Schwarz inequality for the functions 1 and t^2 .

Answer. (i) $|\langle u, v \rangle| = \sqrt{2}$, $\|u\| = \|v\| = \sqrt{2}$; the Cauchy–Schwarz inequality $\sqrt{2} \leq 2$ holds. (ii) $\langle 1, t^2 \rangle = \frac{1}{3}$, $\|1\| = 1$, $\|t^2\| = \frac{1}{\sqrt{5}}$; the Cauchy–Schwarz inequality $\frac{1}{3} \leq \frac{1}{\sqrt{5}}$ holds.

Exercise 7.8 (proof)

- (i) For $u, v \in \mathbb{C}^2$ prove the identity $\|u\|^2 \|v\|^2 - |\langle u, v \rangle|^2 = |u_1 v_2 - u_2 v_1|^2$. Deduce the Cauchy–Schwarz inequality in $\mathbb{C}^2$ and its equality case.
- (ii) Use the Cauchy–Schwarz inequality to prove that for all $a_1, \dots, a_n \in \mathbb{C}$ we have $|\sum_{k=1}^n a_k|^2 \leq n \sum_{k=1}^n |a_k|^2$.

Hint. (i) Expand both sides; the terms $|u_1|^2 |v_1|^2$ and $|u_2|^2 |v_2|^2$ cancel on the left. For the equality case, $u_1 v_2 - u_2 v_1$ is the determinant of the matrix with columns u, v . (ii) Apply the Cauchy–Schwarz inequality on $\mathbb{C}^n$ to $u = (1, 1, \dots, 1)$ and $v = (a_1, \dots, a_n)$, then read off $\|u\|^2$.

Triangle inequality, parallelogram law, and polarization

Exercise 7.9 (computational) Compute $\langle u, v \rangle$, $\|u\|$, $\|v\|$, and the cosine of the angle between the vectors u and v , where (i) $u = (1, 2, 2)$ and $v = (2, 2, 1)$ in $\mathbb{R}^3$; (ii) $u = (1, 1, 1, 1)$ and $v = (1, 1, 1, -1)$ in $\mathbb{R}^4$, where you also find the angle.

Answer. (i) $\langle u, v \rangle = 8$, $\|u\| = \|v\| = 3$, $\cos \theta = \frac{8}{9}$. (ii) $\langle u, v \rangle = 2$, $\|u\| = \|v\| = 2$, $\cos \theta = \frac{1}{2}$, $\theta = 60^\circ$.

Exercise 7.10 (computational) In $\mathbb{C}^2$ take $u = (1, i)$ and $v = (2, 1)$. Use the complex polarization identity to recover $\langle u, v \rangle$ from the four norms $\|v + \alpha u\|^2$ for $\alpha \in \{1, -1, i, -i\}$. Check your result directly.

Answer. $\langle u, v \rangle = 2 - i$; the four squared norms are 11, 3, 5, 9 in the order 1, -1 , i , $-i$.

Hint. First compute the four vectors $v + \alpha u$ and their squared norms; the identity then builds $\langle u, v \rangle$ from these four numbers. Check the result against the formula $\overline{u_1} v_1 + \overline{u_2} v_2$.

Exercise 7.11 (proof) In a *real* inner product space, let $\|u\| = \|v\|$. Prove that $u + v$ and $u - v$ are orthogonal: the diagonals of a rhombus are perpendicular.

Hint. Expand $\langle u + v, u - v \rangle$ by linearity in both slots. Over $\mathbb{R}$ the inner product is symmetric, so the cross terms cancel and only the norms are left.

Exercise 7.12 (★) Prove that the triangle inequality becomes the equality $\|u + v\| = \|u\| + \|v\|$ if and only if one of the vectors u, v is a multiple of the other with a nonnegative real multiplier.

Hint. Square the equality and compare it with the expansion of $\|u + v\|^2$. You reach the equality case of the Cauchy-Schwarz inequality; what remains is to pin down the phase of the multiplier.

Dirac notation

Exercise 7.13 (computational)

- (i) Let $|0\rangle, |1\rangle$ be an orthonormal basis, $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, and $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$. Find the outcome probabilities of a measurement in the basis $|0\rangle, |1\rangle$, and $\langle\sigma_z\rangle$.
- (ii) Answer the same questions for $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$.
- (iii) Take $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Find the probabilities of the outcomes $|0\rangle$ and $|1\rangle$, the amplitude $\langle + | \psi \rangle$, and the probability $|\langle + | \psi \rangle|^2$.

Answer. (i) Probabilities $\frac{9}{25}$ and $\frac{16}{25}$; $\langle\sigma_z\rangle = -\frac{7}{25}$. (ii) Probabilities $\frac{1}{4}$ and $\frac{3}{4}$; $\langle\sigma_z\rangle = -\frac{1}{2}$. (iii) Probabilities $\frac{1}{2}$ and $\frac{1}{2}$; $\langle + | \psi \rangle = \frac{1+i}{2}$, $|\langle + | \psi \rangle|^2 = \frac{1}{2}$.

Exercise 7.14 (computational) In $\mathbb{C}^2$ take $|0\rangle = (1, 0)$, $|1\rangle = (0, 1)$, and $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Write down the matrix of the outer product $|\psi\rangle\langle\psi|$ in the basis $|0\rangle, |1\rangle$.

Answer. $|\psi\rangle\langle\psi| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

Exercise 7.15 (proof)

- (i) Let ϕ, ψ be nonzero and $T = |\psi\rangle\langle\phi|$. Prove that $T^2 = \langle\phi | \psi\rangle T$, and find when $T^2 = T$.
- (ii) Let the vectors $|n\rangle$ form an orthonormal basis, and let $|\psi\rangle = \sum_n c_n |n\rangle$. Prove that $\langle m | \psi \rangle = c_m$ and $\|\psi\|^2 = \sum_n |c_n|^2$.

Answer. (i) $T^2 = T$ exactly when $\langle\phi | \psi\rangle = 1$.

Hint. (i) Apply T^2 to an arbitrary vector $|\chi\rangle$; the number $\langle\phi | \psi\rangle$ comes out of the middle. (ii) For the first equation use $\langle m | n \rangle = \delta_{mn}$. For the second, recall that the ket-to-bra correspondence is conjugate-linear, so $\langle\psi| = \sum_m \overline{c_m} \langle m|$.